

PE-EC603A — Introduction to MEMS

Semester VI • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Introduction and Historical Background

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Fundamentals of MEMS and Microsystems

Micro-Electro-Mechanical Systems (MEMS) represent a technology that combines mechanical elements, sensors, actuators, and electronics on a common silicon substrate through micro-fabrication technology.

1.1 Definition and Concepts

MEMS (Micro-Electro-Mechanical Systems): An integrated micro-scale system combining electrical, mechanical, optical, chemical, or fluidic elements fabricated on a single chip using semiconductor manufacturing processes.

Core Characteristics of MEMS:

- **Scale:** Feature sizes typically range from 1 micrometer (μm) to 100 micrometers (μm), while the overall device footprint is usually between 20 μm and several millimeters.
- **Integration:** Merges high-precision micro-sensors (input), micro-actuators (output), and microelectronic circuitry (control/processing) on a monolithic silicon substrate.
- **Multi-physics:** Involves coupling across mechanical, electrical, thermal, magnetic, optical, fluidic, and chemical domains.

1.2 Microsystems vs. Microelectronics

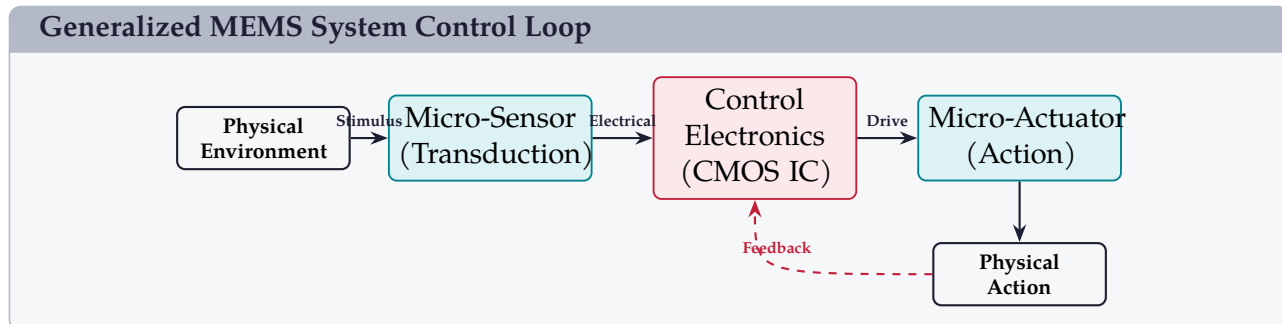
While microelectronics deals with the processing and flow of electrical signals (electrons), microsystems interface directly with the physical environment by converting non-electrical physical variables into electrical signals, and vice versa.

Structural Comparison:

Feature	Microelectronics (IC)	Microsystems (MEMS)
Primary Media	Electrons and electrical signals.	Force, pressure, acceleration, chemical concentrations, heat, light, fluid flow.
Components	Transistors, diodes, resistors, capacitors (static elements).	Cantilevers, diaphragms, micro-valves, comb drives, micro-mirrors (moving parts).
Packaging	Standardized plastic/ceramic packages, sealed from environment.	Must expose the active sensor elements to the environment while shielding the electronics.
Substrate	Purely single-crystal silicon.	Silicon, glass, quartz, polymers (SU-8, PDMS), metals.
Manufacturing	Standard CMOS / planar processes.	CMOS + bulk/surface micromachining, wafer bonding, high-aspect-ratio lithography (LIGA).

1.3 Generalized MEMS System Architecture

A complete microsystem acts as a bridge between the physical environment and the electronic processing domain. It typically contains five functional components:



2. History and Milestones of Miniaturization

The evolution of MEMS spans several decades, transitioning from basic laboratory micro-mechanical discoveries to commercial monolithic consumer electronic devices.

2.1 Richard Feynman's Vision (1959)

The conceptual foundation of nanotechnology and micro-manufacturing was laid by physicist Richard Feynman on December 29, 1959, during his famous American Physical Society lecture at Caltech: *"There's Plenty of Room at the Bottom"*. **Key Concepts Proposed by Feynman:**

- **Writing at Atomic Scale:** Suggested writing the entire 24 volumes of the Encyclopaedia Britannica on the head of a pin by scaling down the font size.
- **Micro-machining Tools:** Envisioned building micro-scale mechanical hands that could build even smaller machines, cascading down to build atomic-scale factories.
- **MEMS Actuators:** Proposed the creation of tiny motors (micro-motors) and electrostatic actuators using micro-lithographic techniques.
- **Physical Scaling:** Highlighted that as machines scale down, the relative importance of physical forces changes (gravity fades, while surface tension and electrostatic forces dominate).

2.2 Key Historical Milestones

The realization of Feynman's vision was enabled by the development of silicon semiconductor manufacturing and micromachining processes:

- **1954 — Discovery of Piezoresistivity in Silicon:** C. S. Smith discovered that silicon exhibits a large change in electrical resistance when mechanical stress is applied. This made silicon an exceptional material for pressure and strain sensors.
- **1967 — Resonant Gate Transistor (RGT):** Developed by Harvey Nathanson at Westinghouse. It was the first batch-fabricated MEMS device, utilizing a moving gold cantilever beam acting as the gate element of a MOS transistor.

- **1970s — Silicon Bulk Micromachining:** Techniques for anisotropic chemical etching of silicon (using KOH) were perfected, enabling the production of thin silicon membranes for pressure sensors.
- **1980s — Surface Micromachining:** R. T. Howe and R. S. Muller at UC Berkeley developed surface micromachining. They used polysilicon as the structural material and silicon dioxide (SiO_2) as the sacrificial material to release free-moving mechanical structures like micro-cantilevers and gears.
- **1989 — Electrostatic Comb Drive:** Developed by W. C. Tang, T. H. Nguyen, and R. T. Howe. It introduced lateral comb structures that generate electrostatic forces, forming the basis of micro-resonators and gyroscopes.
- **1990s — Commercialization and DRIE:** The invention of Deep Reactive Ion Etching (DRIE) by the Bosch Process enabled etching of high-aspect-ratio trenches in silicon. Commercialization succeeded with ADI's ADXL50 accelerometer (for automotive airbag deployment) and TI's Digital Micromirror Device (DMD) for projectors.

3. Applications of MEMS

MEMS technology is ubiquitous across several key industrial sectors:

3.1 Automotive Industry

The automotive sector was the primary driver of MEMS commercialization, requiring high reliability and low-cost sensor components.

- **Airbag Accelerometers:** Capable of measuring sudden deceleration (50 g to 100 g) to deploy airbags within milliseconds.
- **Electronic Stability Control (ESC):** Uses MEMS gyroscopes and accelerometers to measure vehicle yaw rate and lateral acceleration, applying differential braking to prevent skids.
- **Tire Pressure Monitoring Systems (TPMS):** Wireless piezoresistive pressure sensors mounted inside the tires to monitor inflation levels.

3.2 Consumer Electronics

MEMS enabled the miniaturization and touch interface revolution in personal devices.

- **Smartphones:** Utilize 3-axis MEMS accelerometers (screen auto-rotation), gyroscopes (mobile gaming, stabilization), and MEMS barometers (altitude tracking).
- **MEMS Microphones:** Capacitive micro-scale microphones replacing traditional electret condenser microphones (ECM), offering better sound quality, thermal stability, and noise cancellation.
- **Display Systems (TI DMD):** Digital Micromirror Devices contain millions of microscopic electrostatic mirrors on a single chip, controlling light reflections for digital cinema projectors.

3.3 Medical and Biomedical (Bio-MEMS)

The intersection of MEMS and biomedical engineering has created a new class of diagnostics and therapy.

- **Disposable Pressure Sensors:** Piezoresistive catheter tip sensors used for real-time arterial blood pressure monitoring during surgeries.
- **Lab-on-a-Chip (LoC):** Microfluidic channels, pumps, and valves integrated on a chip to perform complex blood analysis and DNA profiling rapidly with microliters of sample.
- **Microneedles and Drug Delivery:** Array of silicon microneedles ($100\text{ }\mu\text{m}$ to $500\text{ }\mu\text{m}$ long) that penetrate the outer dead skin layer painlessly to deliver transdermal drug therapies.

Quick Revision — Module I

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Key Concepts and Parameters

Aspect Ratio (AR)	Aspect Ratio = $\frac{\text{Height } (h)}{\text{Width } (w)}$
Feynman's Scaling Rule	$F \propto L^n$ (Forces scale down differently depending on n)
Piezoresistive Effect	$\frac{\Delta R}{R} = G \cdot \epsilon$ (G = Gauge Factor, ϵ = Mechanical Strain)

One-Line Definitions

- **MEMS:** Micro-scale integrated systems combining mechanical, electrical, and control functions on a single silicon chip.
- **Microsystem:** An engineering system containing sensors, actuators, and electronics to bridge the physical and electrical domains.
- **Aspect Ratio:** The ratio of the height of a microfabricated structure to its lateral width.
- **Piezoresistivity:** The change in electrical resistivity of a material (especially silicon) under applied mechanical stress.
- **Sacrificial Layer:** A temporary film layer (SiO_2) deposited during surface micromachining, etched away later to release moving parts.
- **Structural Layer:** The functional, moving material layer (usually Polysilicon) that forms the core device structure in surface micromachining.
- **DRIE:** Deep Reactive Ion Etching, an anisotropic dry etching process used to fabricate high-aspect-ratio silicon trenches.

Frequently Asked / Viva Questions

Q1. What is the primary difference between a sensor and an actuator in a MEMS device?

A sensor converts physical variables (e.g., pressure, acceleration) into electrical signals. An actuator converts electrical signals into physical motion or action (e.g., micro-mirror tilt, micro-valve opening).

Q2. How did C. S. Smith's 1954 discovery of piezoresistivity influence MEMS technology?

Smith's discovery showed that silicon's resistance changes significantly under stress, forming the transduction basis for pressure sensors, accelerometers, and strain gauges.

Q3. Why did Richard Feynman's 1959 speech serve as the conceptual basis for MEMS?

Frequently Asked / Viva Questions (contd.)

Feynman predicted that machines could be scaled down, that micro-machining could build micro-motors, and that physical scaling would shift dominance from gravity to electrostatic and surface forces.

Q4. What is the structural role of a sacrificial layer in surface micromachining?

It acts as a temporary spacer supporting the structural layer during deposition. Once etched away (released) with an acid, it leaves the structural layer suspended and free to move.

Q5. Why is single-crystal silicon the dominant substrate material in MEMS?

Silicon has excellent mechanical properties: no mechanical hysteresis, high fatigue strength, can be easily batch-fabricated using established CMOS semiconductor facilities.

Q6. What was the Resonant Gate Transistor (RGT) and why was it a milestone?

Developed in 1967, it was the first batch-fabricated MEMS device. It used a moving gold cantilever as a transistor gate, proving mechanical elements could be integrated with microelectronics.

Q7. Why is packaging more complex for MEMS devices compared to standard Integrated Circuits (ICs)?

ICs are hermetically sealed from the environment. MEMS sensors must interface directly with external media (like air or fluid) while simultaneously protecting sensitive control electronics.

Q8. Explain the function of a MEMS gyroscope in consumer electronics.

It measures the angular rate of rotation (yaw, pitch, roll), enabling camera image stabilization, vehicular electronic stability control, and smartphone motion controls.

Q9. What is LIGA technology and when is it preferred?

LIGA is a German acronym for lithography, electroplating, and molding. It uses high-energy X-ray lithography to fabricate structures with extremely high aspect ratios.

Q10. State two applications of Bio-MEMS.

Disposable catheter tip blood pressure sensors and Lab-on-a-Chip (LoC) microsystems for rapid blood diagnosis and DNA sequencing.

Q11. How does a MEMS accelerometer in an automotive airbag sensor function?

A suspended proof mass deflects under sudden deceleration, changing the capacitance between moving and fixed electrodes to trigger the airbag within milliseconds.

Comparison: Silicon (MEMS) vs. Steel (Macro-engineering)

Property	Silicon (Si)	Structural Steel
Yield Strength	7 GPa (higher than steel).	0.2 – 0.5 GPa.
Young's Modulus	190 GPa (stiff, similar to steel).	200 GPa.
Mechanical Hysteresis	Negligible (extremely elastic).	Significant (plastic deformation).
Fatigue Lifetime	Infinite (no dislocation movement under stress).	Finite (dislocations accumulate, causing fracture).
Density	2.3 g/cm ³ (lightweight).	7.8 g/cm ³ .